

Sayali Sanjay Solanke

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Research Interest

Exploring molecular mechanisms of disease, with an interest in applying chemistry to understand biological processes and develop new therapeutic approaches.

Education

IISER Thiruvananthapuram, BS–MS Integrated Dual Degree 2022 – 2027

- Major: Chemistry | Minor: Biology

Keshawraj Junior College, Latur, Maharashtra 2019 – 2021

- Higher Secondary Education

Research Experience

Master's Thesis - Main Group Chemistry and Metal-Ligand Reactivity, Aug 2026 – Present
IISER Tvm

Supervisor: Dr. Milan Bisai

- Working on the synthesis and study of main group metal complexes and their ligand systems.
- Synthesizing NacNac ligands and its derivatives as supporting ligands for main group metal chemistry.
- Developing experience in air sensitive synthesis, including Schlenk line techniques, solvent handling, reaction setup, and purification.
- Characterizing synthesized compounds using spectroscopic and analytical techniques and studying their structural and coordination properties.
- Exploring the role of ligand design in controlling the structure and reactivity of main group metal complexes.

Outcome: Developing experience in inorganic ligand synthesis and handling.

Topochemical Polymerization Project, IISER Thiruvananthapuram Aug 2024 – Aug 2026

Supervisor: Prof. Kana M. Sureshan

- Synthesized functionalized monomers through organic transformations including amide coupling, Boc deprotection, esterification, and acylation reactions.
- Purified and characterized synthesized intermediates and monomers using chromatographic and spectroscopic techniques.
- Characterized monomers using NMR spectroscopy, PXRD, and thermal analysis techniques including DSC and TGA.
- Studied the relationship between monomer crystal packing, molecular structure, and polymerization behaviour.

Outcome: Synthesized and characterized functional monomers intended for topochemical polymerization; developed expertise in multi-step organic synthesis, reaction work-up, purification, and spectroscopic characterization.

Seed and Starch Biology Research Project, IISER Thiruvananthapuram

Supervisor: Dr. Nitin Uttam Kamble

- Applied solid-state CP/MAS NMR spectroscopy to investigate the structural characteristics of starch derived from different seed systems like chickpea, Nipponbare Rice, Pusa Basmati.
- Investigated molecular changes associated with seed aging and stress response using rice seed systems, including wild-type, OsPIMT1, and OsPIMT2 lines.
- Performed solution-state ^1H NMR analysis to compare metabolite profiles of wild-type and PIMT-overexpressing seed samples under control and aging conditions.
- Analyzed NMR spectral features to identify differences in metabolite composition and carbohydrate characteristics associated with PIMT expression and seed aging.

Outcome: Characterized molecular changes in rice seed metabolites associated with PIMT overexpression and aging using solution state NMR, and investigated structural changes in starch containing seeds using solid-state NMR.

Winter Research Project – Computational Materials Physics

Dec 2025

Supervisor: Dr. Bhalchandra Pujari

- Completed coursework in computational materials science (Ghent University).
- Basic theoretical analysis of electronic and structural properties of solids.

Summer Research Internship – Parasitology, Dayanand Science College

Jun – Jul 2024

Supervisor: Dr. Amol Shankarrao Thosar

- Parasitological analysis of cestodes (sheep) and nematodes (fish).
- Dissection, staining, microscopy, and pathology assessment.

Technical Skills

Experimental Techniques: Organic synthesis; UV-Vis, IR, and NMR spectroscopy; X-ray diffraction; thermogravimetric analysis; differential scanning calorimetry; cyclic voltammetry; polarimetry; conductometry; plant DNA extraction; gel electrophoresis; basic cell culture techniques

Software: TopSpin, ChemDraw, OriginPro, Mercury, CCDC

References

Available on request